

Risk Assessment Form

School(s)/ Service(s):	School of Social Sciences - Psychology	Manager Responsible:	Michelle Dalton
Risk Assessment Title:	Use of Polar H10 Heart Rate Monitor (chest-strap) in research and teaching	Assessment No:	101
Location of Activity:	Psychology laboratories and approved field/off-site testing locations	Name of Assessor:	Caspar Wynne
Details of Activity:	Fitting and use of the Polar H10 chest-strap heart rate sensor on research participants and students during psychophysiology studies and teaching. The device is a Bluetooth chest-worn sensor held against the skin below the chest by an elasticated strap moistened with water. It records heart rate and inter-beat interval data, which is streamed to a paired phone, tablet or computer. Participants are briefed, give informed consent, and fit the strap themselves in private wherever possible. The activity covers device fitting, data collection, cleaning and storage.		
Other assessments or documents which might also be required, X if needed:			
Manual Handling ☐ COSHH ☐ Noise ☐ Other (please specify) ☐ Click or tap here to enter text.			

Signature of Assessor	
Signature: Caspar Wynne	Date: 10/08/2026
Signature of Manager(s) "The risks identified in this assessment are controlled so far as is reasonably practicable"	
Signature: Click or tap here to enter text.	Date: Click or tap to enter a date.

Date of Reassessment (Every two years minimum)	Are There Any Changes To The Activity Since The Last Assessment?	Signature of Manager
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap to enter a date.
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap to enter a date.
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap to enter a date.

Name of Person Undertaking the Activity	School/ Service	Role	Signature	Date
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap to enter a date.
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap to enter a date.
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap to enter a date.

LIKELIHOOD (L)	
5	Inevitable
4	Highly Likely
3	Possible
2	Unlikely
1	Remote Possibility

SEVERITY (S)	
5	Very High –Death or permanent disability
4	High – Serious injury (hospital admission)
3	Moderate - RIDDOR over 7 days
2	Slight - First Aid treatment
1	Nil - Very Minor

RISK RATING	ACTION
1 – 4	Broadly Acceptable - No action required
5 – 9	Moderate - Reduce risks if reasonably practicable
10 – 15	High Risk - Priority Action to be undertaken
16 – 25	Unacceptable - Action must be taken IMMEDIATELY

RISK RATING = LIKELIHOOD X SEVERITY						
SEVERITY (S)	5	5	10	15	20	25
	4	4	8	12	16	20
	3	3	6	9	12	15
	2	2	4	6	8	10
	1	1	2	3	4	5
		1	2	3	4	5
		LIKELIHOOD (L)				

HAZARD e.g. Falling Objects, Fire, Explosion, Noise, Violence etc.	PERSONS AT RISK e.g. Employees, Contractors, Members of the public	POSSIBLE OUTCOME	RISK RATING WITHOUT CONTROLS (LXS)	CONTROL MEASURES e.g. Guards, Safe Systems of Work, Training, Instruction, Authorised Users, Competent Persons, Personal Protective Equipment (PPE)	RISK RATING WITH CONTROLS (LXS)	FURTHER ACTION REQUIRED? Yes/No
Skin contact / hygiene – shared chest strap worn against skin by multiple users	Participants, students, staff	Cross-contamination or transfer of skin infection between users; minor skin irritation.	3x2 = 6	Strap cleaned/disinfected with anti-bacterial wipes between every user and washed with warm soapy water after each session per manufacturer guidance. Single-user allocation where possible. Users with broken/irritated skin excluded. Hand hygiene available. Straps stored clean and dry.	1x2 = 2	No
Skin irritation / allergic reaction – elastic strap, moisture and prolonged contact	Participants, students	Redness, itching, contact dermatitis or allergic reaction to strap material or moisture.	2x2 = 4	Participants screened for known latex/elastic/textile allergies and sensitive-skin conditions before fitting; excluded if sensitive. Strap fitted only as long as needed and removed promptly. Participants told to report any discomfort and to remove the strap immediately if irritation occurs.	1x2 = 2	No
Fitting on/near the chest – discomfort, embarrassment or inappropriate contact during fitting	Participants, students	Psychological discomfort/distress; safeguarding concern if fitting not managed appropriately.	3x2 = 6	Participants always fit the strap themselves in a private space (screen/changing area). Researcher gives verbal/diagram instructions and does not fit the strap on the participant. Informed consent obtained; right to withdraw made clear. Chaperone available on request. Under-18s only with appropriate ethics approval and safeguarding measures.	1x2 = 2	No

Use on individuals with cardiac implants / pacemakers or metal chest implants	Participants	Interference concern / unsuitable use; participant distress.	2x3 = 6	Pre-participation screening question excludes anyone with a pacemaker, implanted cardiac device or metal implant in the chest area. Device used for research/teaching only, not for medical diagnosis. Manufacturer guidance followed.	1x3 = 3	No
Misinterpretation of heart-rate data as a medical reading; undetected health issue	Participants	Participant alarm or false reassurance; failure to signpost a genuine health concern.	2x3 = 6	Participants clearly told the device is for research/teaching, not a medical diagnostic tool. Staff do not give medical interpretation or advice. If a reading or symptom causes concern, standard first-aid/emergency procedures followed and participant advised to seek medical advice.	1x2 = 2	No
Physical activity studies – exertion while wearing the monitor	Participants	Faint, fall, over-exertion, pre-existing condition aggravated during exercise task.	3x3 = 9	Physical-activity protocols use their own separate exercise/PARQ+ risk assessment and health screening. Water available; participants told to stop if unwell. Trained staff present; first-aid cover and emergency procedures in place. Device is water-resistant and rated for activity use.	1x3 = 3	Yes
Hygiene of moistening the strap electrodes (shared water/contact)	Participants, staff	Minor cross-contamination via shared water source.	2x1 = 2	Strap moistened by the individual user under running tap water or with their own water; no shared standing water. Strap disinfected between users as above.	1x1 = 1	No

Data protection – heart-rate/physiological data is personal data (special category if health-related)	Participants	Breach of GDPR; identifiable personal/health data disclosed.	3x3 = 9	Data collected, stored and processed per University data-protection policy and study ethics approval. Data pseudonymised, stored on approved encrypted University systems, app accounts secured, participant consent for data use obtained. Data retained/deleted per ethics protocol.	1x3 = 3	No
Small parts / strap – damage, wear, or ligature-style misuse in vulnerable groups	Participants (esp. vulnerable groups)	Injury from damaged device; misuse risk with at-risk participants.	1x3 = 3	Straps and sensors inspected before each use and withdrawn if worn/damaged. Devices signed out and supervised during sessions. Additional safeguarding controls applied for vulnerable participant groups per ethics approval.	1x2 = 2	No
Battery / electrical (coin-cell battery in sensor)	Staff, participants	Battery leak; ingestion risk of coin cell if removed; minor.	1x3 = 3	Battery compartment kept closed; battery changed only by trained staff and spares stored securely away from participants (coin-cell ingestion hazard). Damaged units removed from use.	1x2 = 2	No
Hygiene during infectious-disease periods (e.g. respiratory illness outbreaks)	Participants, staff	Increased transmission risk of infection via shared equipment.	3x2 = 6	Enhanced cleaning regime and any current University/public-health guidance followed during outbreak periods; single-user allocation or a quarantine period between users; use suspended if guidance requires.	1x2 = 2	No
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.

Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.

ACTION If further action has been identified above, describe what needs to be done, by whom with agreed timescales for completion			
Description	Who	Target Date	Completed On
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap to enter a date.	Click or tap to enter a date.
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap to enter a date.	Click or tap to enter a date.
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap to enter a date.	Click or tap to enter a date.

COMMENTS Use this section to record any additional information, comments, dynamic risk assessment comments etc.
Click or tap here to enter text.